SUPERSTORE SALES DASHBOARD

Week 2 Report: Data Modeling, DAX & Sales Dashboard

Name:Kamali V

Table of Contents:

1. Problem Statement & Objectives
2. Data Preparation Recap (Power Query)
3. Data Model
4. DAX Measures & Report Build
5. Interactivity: Slicers, Cross-Filtering & Tooltips
6. Final Dashboard & Conclusion

1. Problem Statement & Objectives:

Management wants an interactive dashboard to monitor sales, profit, orders, customers, and regional performance. This report builds on the cleaned Superstore dataset from Week 1 and documents the data model, DAX measures, and visuals used to deliver the Superstore Sales Dashboard.

Objectives:

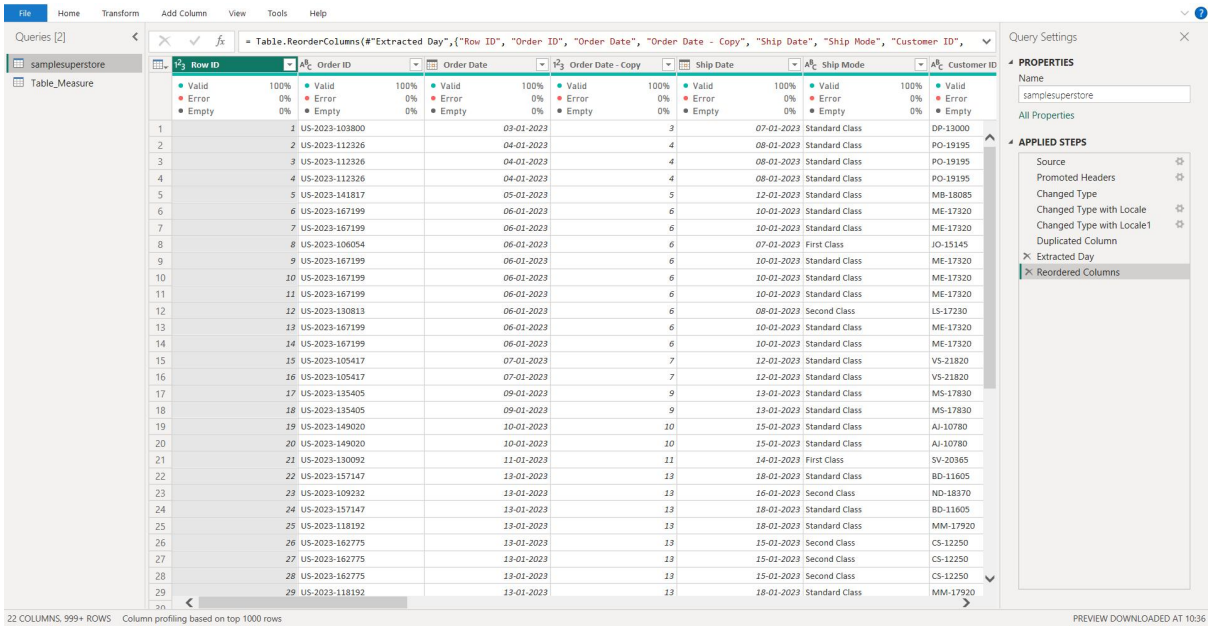
- Establish table relationships and data model structures (Star Schema).
- Write foundational DAX measures: Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales, and Profit Margin %.
- Construct visual layouts using KPI cards, stacked bar charts, line charts, and decomposition trees.
- Configure slicers, cross-filtering, interactive tooltips, and drill-down paths.

Tasks & Deliverables:

- Build a dedicated DAX measure table (_Measures).
- Create a top-line executive summary bar with key KPI cards.
- Build category performance and regional profitability visual breakdowns.
- Deliver an interactive Superstore Sales Dashboard.

2. Data Preparation Recap (Power Query):

The Week 1 preprocessing in Power Query Editor is the foundation for the model. The query samplesuperstore contains 22 columns, and column profiling shows every visible column is 100% valid with 0% errors and 0% empty values.

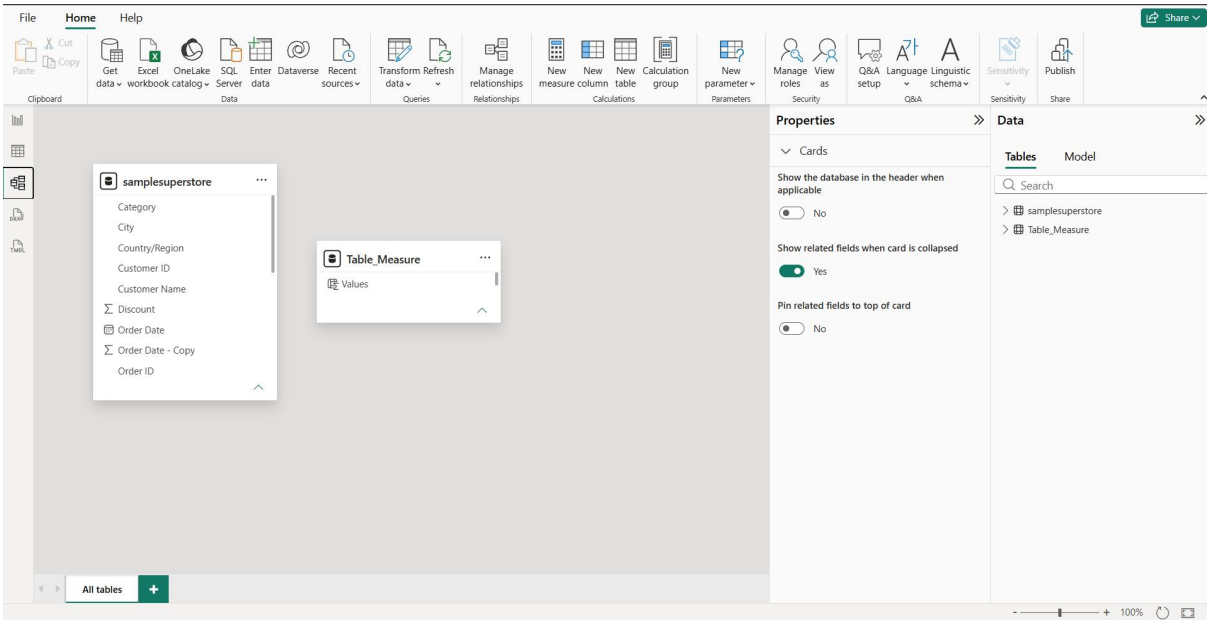


Key Points:

- **Type coercion:** Source, Promoted Headers, Changed Type, and two Changed Type with Locale steps set correct data types, including dd-mm-yyyy dates.
- **Temporal attributes:** Order Date was duplicated (Order Date - Copy) and the Day was extracted, giving a numeric day field for the line chart.
- **Column order:** Reordered Columns places related fields (Row ID, Order ID, dates, Ship Mode, Customer ID) together for easier navigation.
- **Measure table:** A second query, Table_Measure, is loaded to hold the DAX measures.

3. Data Model:

Model view in Power BI Desktop shows how the loaded tables are organised. The model contains the main fact table, samplesuperstore, and the dedicated measure table, Table_Measure.

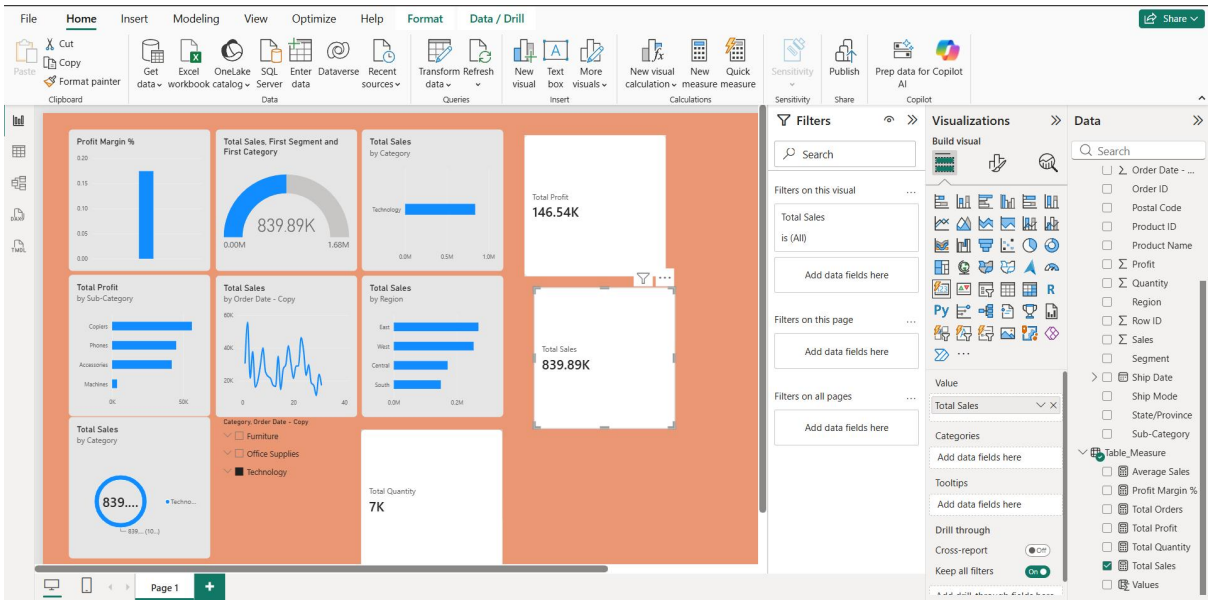


Key Points:

- **Fact table:** samplesuperstore holds the transactional and descriptive fields: Category, City, Country/Region, Customer ID, Customer Name, Discount, Order Date, Order Date - Copy, Order ID, and more.
- **Measure table:** Table_Measure has no business columns of its own (only a placeholder Values column). It exists to keep all DAX measures in one organised place.
- **Relationships:** Because Table_Measure only stores calculations, it does not need a relationship line to the data table; measures reference samplesuperstore directly.
- **Next step:** To extend towards a full star schema, separate Date, Product, Customer, and Region dimension tables could be created and related to the fact table using Manage relationships.

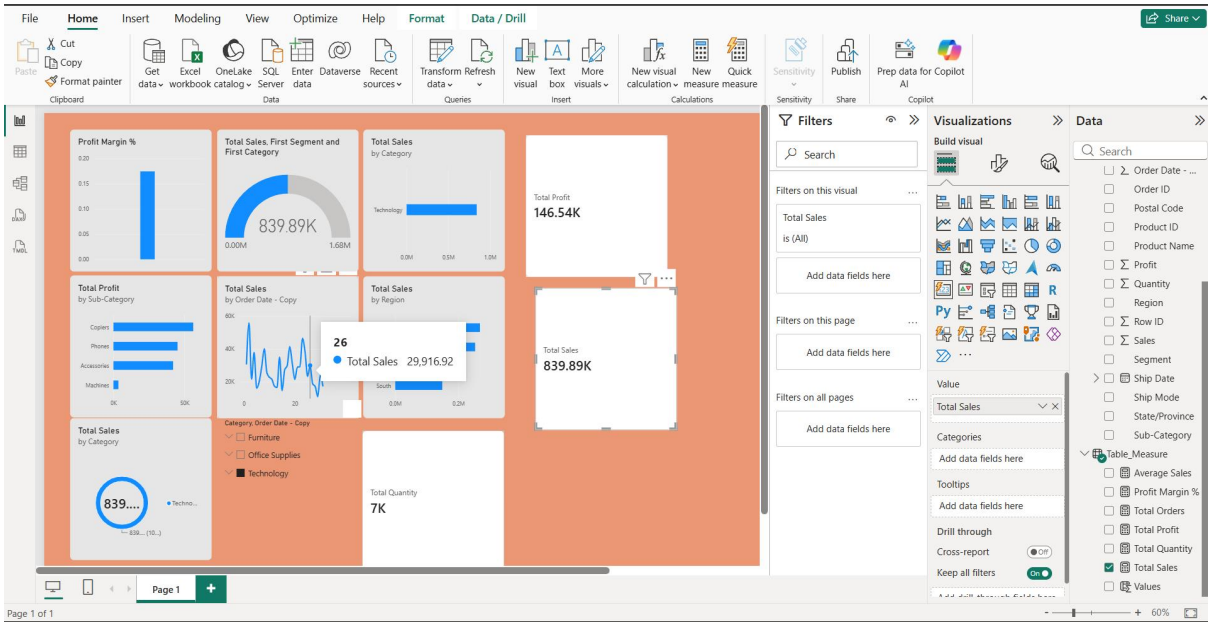
4. DAX Measures & Report Build:

All foundational measures were created in Table_Measure and are available in the Data pane. They power the KPI cards, gauge, and charts on the report canvas.



5. Interactivity: Slicers, Cross-Filtering & Tooltips:

The dashboard is designed so users can explore the data rather than read static numbers. Selecting a category or hovering over a data point updates or explains the other visuals on the page.

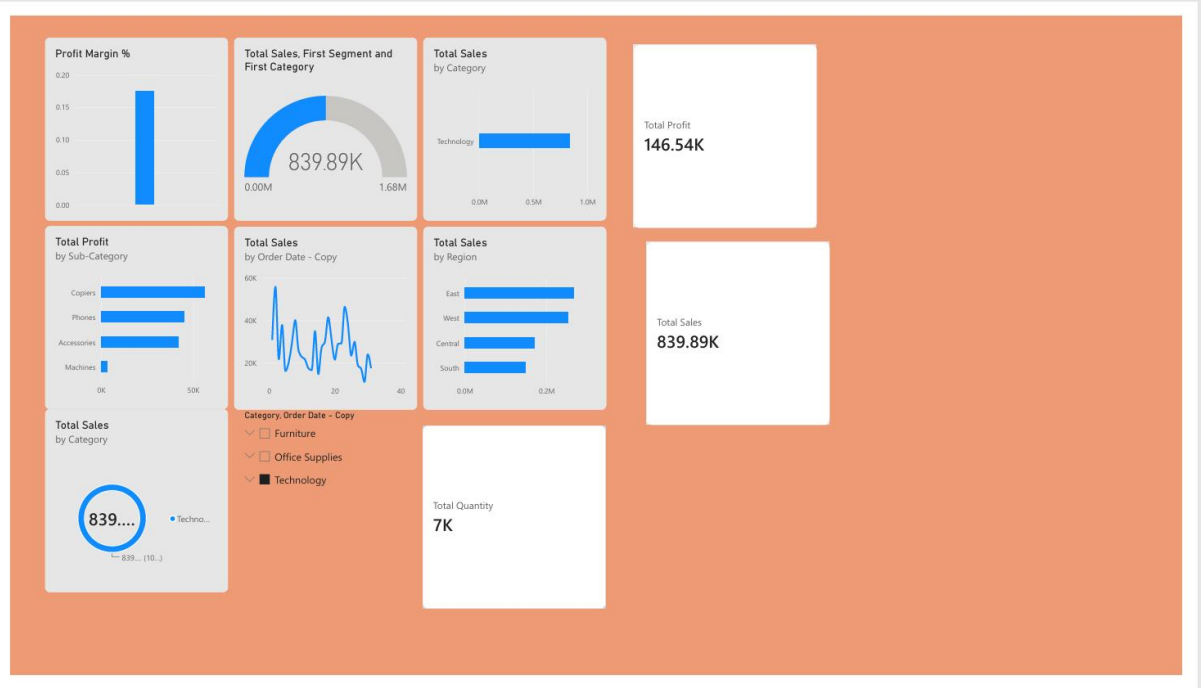


Key Points:

- **Tooltips:** Hovering on the line chart displays a tooltip with the exact value (for example, Total Sales of 29,916.92 for day 26).
- **Slicer & cross-filtering:** The category selector (Furniture, Office Supplies, Technology) filters the KPI cards, gauge, donut, and bar charts together.
- **Filters pane:** The Filters pane supports visual-level, page-level, and all-pages filters, and Keep all filters is switched On for drill-through.
- **Drill-down:** Total Sales by Category, Sub-Category, and Region allow users to drill from a high-level view into finer detail.

6.Final Dashboard & Conclusion:

The finished Superstore Sales Dashboard combines executive KPI cards with category, sub-category, regional, and time-based breakdowns on a single interactive page.



Dashboard Summary:

- **KPI cards:** Total Sales 839.89K, Total Profit 146.54K, and Total Quantity 7K, as shown for the current selection (Technology).
- **Gauge & margin:** The gauge shows Total Sales against a 1.68M scale; the profit margin visual is scaled to 0.20.
- **Breakdowns:** Total Profit by Sub-Category shows Copiers, Phones, and Accessories leading and Machines lowest; Total Sales by Region shows East and West ahead of Central and South.

Conclusion:

Week 2 moved the cleaned Superstore data into a working analytical solution: a measure table with six core DAX measures, a set of KPI and chart visuals, and interactive filtering that lets management monitor sales, profit, orders, and regional performance.